



Smart Glass That Can Read Books

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We have seen many smart glasses to detect the distance between objects or help visually challenged persons to recognise people's faces. Here we extend it to the next level using optical character recognition (OCR). Expanding the DIY project to its next version, we are now giving details to make a smart glass that can read text using OCR and computer vision. A camera mounted on the front of the glass for OCR and computer vision will thus allow visually challenged people also to read. In this version, you will also need a pair of goggles.

Prerequisites

First, we need to install some of the required modules and libraries for the project. As the smart glass is based on real-time video processing, by using OCR we can extract the text from the pages that may include images. Here, a module is needed that can capture images from a camera. By using a speech synthesis module, the text in those captured images can be converted into speech.

First, we need to install the following libraries:

- OpenCV
- PyTesseract
- eSpeak

To install them, use the commands below:

```
sudo apt-get install espeak
sudo apt-get install espeak python-espeak
sudo pip3 install opencv2
```

Coding

Import the required libraries to the code and then set the path where the video frames are to be saved for text extraction.

Create a while loop in the code, which will capture real-time video from the camera. Using cv2, convert the image into BGR and save it to

```
import cv2
import numpy as np
import time
from espeak import espeak
import datetime
import pytesseract
img_cv = cv2.imread(r'/home/pi/pytesseract/tests/data/test.png')

previous = "unkno"

video_capture = cv2.VideoCapture(0)

process_this_frame = True
```

Fig. 1: Code snippet importing Python modules

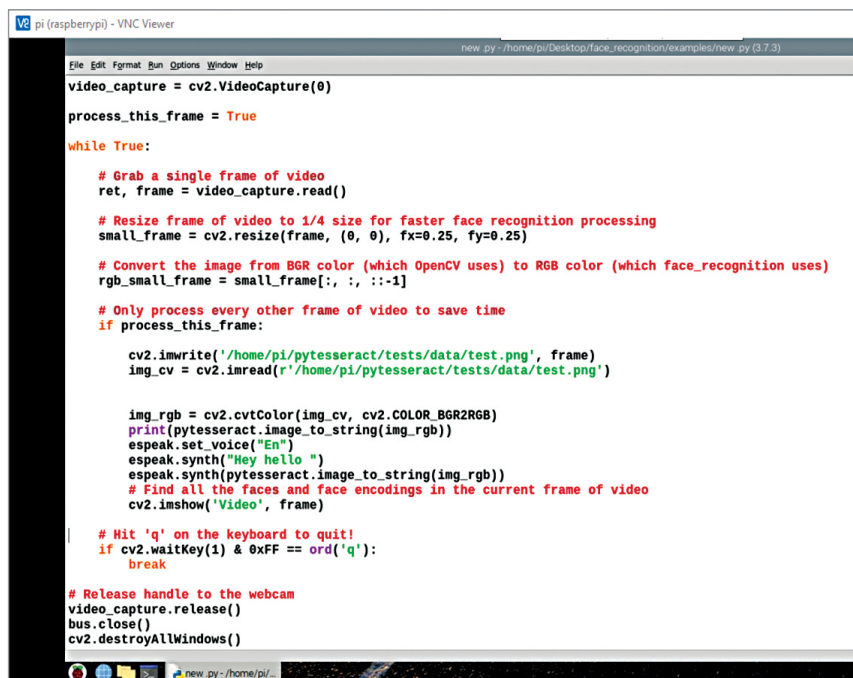


Fig. 2: Code snippet for capturing video and processing it for OCR

EFY Note
The source code
of this project
is available
for free download
at source.efymag.com

Components

Table 1

Component name	Quantity	Description
Raspberry Pi 4	4	2GB/4GB
5V power bank	1	Powering
RPI camera	1	Camera module for RPi
Pam amplifier	1	Amplifier module
Speaker	1	Small speaker